

Erik Bloomquist

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EDUCATION

Bachelor of Science in Electrical Engineering
University of Florida

August 2022 - May 2026
GPA: 3.98/4

WORK EXPERIENCE

Undergraduate Teaching Assistant | *UF ECE Dept.*

January 2026 - Present

- Assists Dr. Catia Silva in operation of "Applied Machine Learning Systems" course
- Course topics include Data Processing and Experimental Design, implementation of various regression, classification, clustering, and dimensionality reduction algorithms with Scikit-Learn, and implementation of various deep learning architectures (MLPs, CNNs, Autoencoders, RNNs, and Transformers) using Pytorch

Undergraduate Teaching Assistant | *UF ECE Dept.*

August 2024 - December 2025

- Assisted Dr. Catia Silva in operation of "Fundamentals of Machine Learning" course
- Course topics include Bayesian learning, generative models, clustering, discriminative classification, dimensionality reduction, manifold learning, neural networks, and deep learning

Undergraduate Teaching Assistant | *UF Engineering Education Dept.*

May 2023 - July 2024

- Assisted Dr. Lilianny Virguez in operation of Elements of Electrical Engineering
- Course topics include AC & DC circuit analysis, Arduino projects, and basic C++ programming

Student Assistant- Database | *UF College of Dentistry*

May 2023 - September 2023

- Worked with the UF College of Dentistry to migrate 5000 medical records into the College's database

RESEARCH EXPERIENCE

Spectral Feature Extraction for Reinforcement Learning* | *Dr. Catia Silva*

- Designing reinforcement learning framework for financial market prediction using spectral features; serving as project leader and sole student author
- Work supported by grant funding from UF Center for Undergraduate Research's AI Scholars Program

Literature Review: Adversarial Machine Learning | *Dr. Damon Woodard*

- Surveying adversarial ML attacks and defenses to assess effectiveness and guide future research in robust deep learning
- Work supported by the Florida Institute for National Security (FINS)

From Black-Box to Glass-Box Models: Towards Explainability in Reinforcement Learning | *Dr. Damon Woodard*

- Trained common RL algorithms (DQN, PPO) on benchmark environments (Cart Pole, Mountain Car, Lunar Lander)
- Examined decision boundaries and extracted interpretable policy representations

AWARDS

Dr. Joseph S. Rosko Award | *Scholarship*

Fall 2025

- Awarded \$5000 scholarship for academic achievement and involvement in the ECE department

AI Scholars Program | *Research Funding*

Fall 2025

- Awarded \$1750 grant to lead independent research bridging machine learning and financial markets

Florida Institute for National Security (FINS) Talent Pipeline | *Fellowship*

Spring 2024

- Awarded \$5000 fellowship to perform research at the intersection of AI and national security

University Honors Program | *Recognition*

Fall 2022

- Admitted to UF's University Honors Program, providing access to advanced coursework and academic enrichment

PRESENTATIONS

2026 Spring Undergraduate Research Symposium* | *Poster Presentation*

Spring 2026

- Will present my AI Scholars Program work on self-directed research in machine learning for financial market prediction

2025 Spring Undergraduate Research Symposium | *Poster Presentation*

Spring 2025

- Accepted to present poster on interpretable reinforcement learning at the 2025 UF Spring Symposium.

LEADERSHIP & INVOLVEMENT

Undergraduate Curriculum Chair | *Eta Kappa Nu (HKN) Epsilon Sigma* Spring 2024 - Present

- Represents IEEE’s honor society at faculty curriculum committee meetings
- Selected to serve on the ECE student panel, answering curriculum-related questions from underclassmen in electrical and computer engineering

PROJECTS

Spectral Feature-Based Stock Trading Agent

- Designed and implemented a reinforcement learning framework for financial market prediction using spectral features. Documented the impact of spectral penalty terms in the reward function on agent performance.

ECG Anomaly Detection Models

- Developed several machine learning pipelines for real-time detection of cardiac anomalies using the MIT-BIH Arrhythmia Database. Compared classical ML algorithms and deep learning models, with emphasis on feature extraction, signal preprocessing, and classification performance.

Autonomous Tabletop Poker-Playing System

- Designed and created an embedded, computer-vision–driven platform built around a Raspberry Pi that captures images of the table, identifies cards and game state, and uses decision logic to determine optimal actions. It integrates custom hardware (controllers, cameras, and analog circuitry) with software for image processing and strategy, enabling semi-autonomous or autonomous play in a physical poker environment.

Blackjack Game System

- Designed and implemented an embedded blackjack game using a Raspberry Pi Pico, integrating an I2C character LCD, PIR sensor, push-button inputs, and DAC-driven audio output. Built custom PCB with regulated power rails, signal conditioning, and an LM386-based amplifier.

SKILLS

Programming & Tools: Python (PyTorch, TensorFlow, scikit-learn, NumPy, Pandas), MATLAB, C++
Data & Tools: Git, Altium Designer, LTspice, LaTeX, Jupyter, SQL
Machine Learning: Deep Learning, Reinforcement Learning, Adversarial ML, End-to-End ML Systems, Data Cleaning & Preprocessing
Signals & Systems: Digital Signal Processing, Biomedical Signal Analysis, Circuit Modeling
Professional Skills: Technical Writing, Effective Communication, Poster Presentations, Teaching & Mentorship

RELEVANT COURSEWORK

Machine Learning Mathematics	Applied Machine Learning Systems
Formal Methods for Robotics & AI	Data Science
Stochastic Methods	Communication Systems
Digital Signal Processing	Biomedical Image Analysis
Analog Circuitry	Electrophysics